W05

GIS fundamentals in QGIS

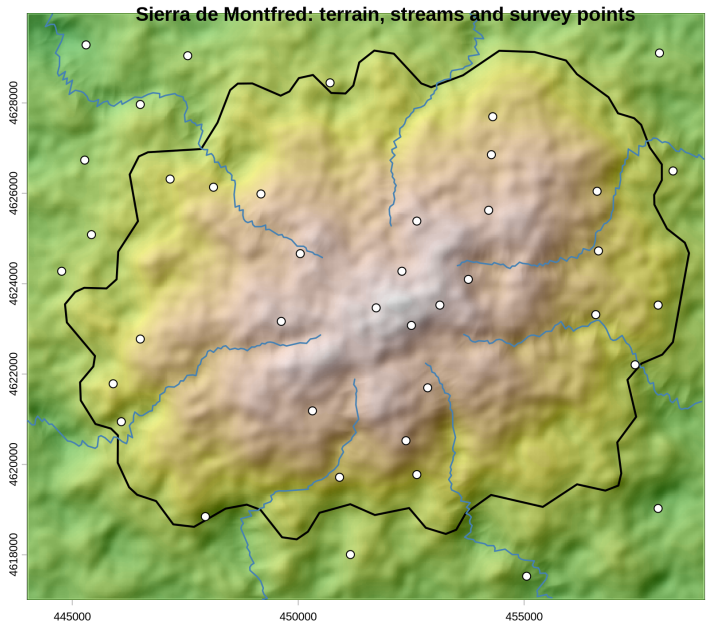
GIS & MAPPING

This week is all QGIS. You work with the Sierra de Montfred data as in a real project: vector and raster layers, coordinate reference systems, buffers, clipping and extracting values. Week 6 repeats the same workflow in R.

DURATION	About 2 hours.
YOU NEED	QGIS LTR installed.
FILES	`montfred.gpkg`, `dem.tif`, `landcover.tif`, `landcover_classes.csv`, `survey_points.csv`
DATA	A fictitious natural park of about 11,000 ha: boundary, streams, survey points, a 30 m digital elevation model (DEM) and land cover.

Learning outcomes

- tell vector from raster data and know when to use each;
- understand coordinate reference systems (CRS) and reproject layers;
- style a DEM with a hillshade and a land-cover map by category;
- run buffers, clips, select-by-location and extract raster values at points;
- calculate areas per class inside the park.



What you will build: hillshaded terrain, park boundary, streams and survey points (figure made in R in Week 6).

1 · Vector, raster and reference systems

	Vector	Raster
Represents	Discrete objects: points, lines, polygons	Continuous surfaces on a grid of cells
Examples here	Survey points, streams, park boundary	Elevation (DEM), land cover
Formats	GeoPackage (.gpkg), shapefile	GeoTIFF (.tif)

A **CRS** defines how coordinates translate into positions on Earth. The points come in **EPSG:4326** (WGS 84, degrees of latitude and longitude), fine for GPS but not for measuring. The other layers are in **EPSG:25831** (ETRS89 / UTM zone 31N), in metres, the official system in eastern Spain. To measure distances and areas, every layer must be in a projected CRS in metres.

The project CRS and each layer's CRS

QGIS reprojects "on the fly": it draws layers in different CRSs together using the project CRS (bottom right). But analysis tools use each layer's own CRS, so we save the points in EPSG:25831 before measuring anything.

2 · Load and style

1. **Project > Properties > CRS**: search 25831 and pick *ETRS89 / UTM zone 31N*. OK.
2. Drag `montfred.gpkg` from the **Browser** panel (or **Layer > Add Layer > Add Vector Layer**) and select park_boundary, streams and survey_points.
3. Add `dem.tif` and `landcover.tif` with **Layer > Add Layer > Add Raster Layer**.
4. DEM: Properties > Symbology > **Singleband pseudocolor**, any colour ramp you like.
5. Hillshade: **Raster > Analysis > Hillshade** on the DEM (azimuth 315°, altitude 45°). Put the DEM on top with **Blending mode: Multiply** in its symbology.
6. Land cover: Symbology > **Paletted/Unique values** > **Classify**. Replace labels 1–8 with the names in `landcover_classes.csv`.
7. Park boundary: transparent fill and a thick black outline. Streams: a blue line.

Code	Class	Code	Class
1	holm oak forest	5	heathland
2	mixed forest	6	meadow
3	pine plantation	7	cropland
4	beech forest	8	urban

3 · Analysis

Park area

1. Open the park_boundary attribute table, toggle editing and open the **Field Calculator**. Create `area_ha` (decimal) with the expression `$area / 10000`.
2. Compare it with the area_ha field the layer already has. Save and stop editing.

Riparian zone: 150 m buffer

1. **Vector > Geoprocessing Tools > Buffer**: layer streams, distance 150 metres, tick **Dissolve result**. Save as riparian.gpkg.
2. **Vector > Research Tools > Select by Location**: features of survey_points that *intersect* riparian. How many points are riparian?

Clip the raster to the park

1. **Raster > Extraction > Clip Raster by Mask Layer**: input landcover, mask park_boundary. Save as landcover_park.tif.
2. In the **Processing Toolbox**, run **Raster layer unique values report** on landcover_park. It gives the cell count and area of each class.

Raster values at the points

1. Processing Toolbox > **Sample raster values**: points survey_points, raster dem, prefix `dem_`. Repeat with landcover.
2. Open the result's table and compare dem_1 with elevation_m (the field GPS reading). Why do they not match exactly?

> **ArcGIS Pro**
>
> The equivalent tools are under **Analysis > Tools**: *Buffer*, *Select Layer By Location*, *Extract by Mask* and *Extract Values to Points* (the last two need the Spatial Analyst extension). Change the CRS in **Map Properties > Coordinate Systems**.

Exercises

1. Add survey_points.csv as delimited text (EPSG:4326) and save it as a GeoPackage in EPSG:25831 (**Export > Save Features As**, changing the CRS). Use the **Measure Line** tool (Ctrl + Shift + M) to measure the distance between P01 and P02.
2. Calculate slope (**Raster > Analysis > Slope**) and extract its value at the points.
3. What percentage of the park is beech forest? And meadow?
4. Reclassify the DEM into elevation belts (< 600, 600–1000, 1000–1300, > 1300 m) with **Reclassify by table**.

5 Save the project as week05.qgz. We reuse it to make maps in Week 7.

Before Week 6

- Install `install.packages(c("sf", "terra"))`. On macOS or Linux, tell me if it fails.
- Write down your exercise results: in Week 6 we repeat them in R and compare.